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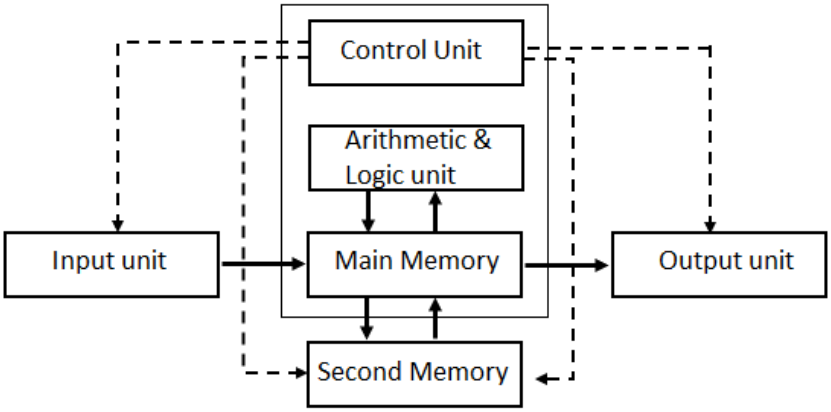
Lesson 1: Basic concepts of computer system

Computer is an electronic device that is designed to perform logical and arithmetic operations sequentially. It takes input (Data) from users via input devices and gives the desired output (Information) via output devices.

Computer Organization: Figure given below shows the block diagram of functional unit computer system:



- **Input unit:** Input unit is used to enter raw data or information into computer system for processing.
- **Storage unit:** Storage unit is used to store input data, intermediate data while processing and output data after processing.
- **Processing unit:** Processing unit performs all the logical and arithmetic operations on the data provided by the user. It consists of Arithmetic Logic Unit (ALU): It performs all the arithmetic functions and Control Unit (CU): takes care of all the operations performed.
- **Output unit:** Output unit is used to store the result provided by computer system after processing.

Computer Memory: Any physical device capable of storing data or information in it either temporary or permanent is known as Computer memory. Computer memory can be divided into many categories:

a) Primary or Main memory: It is directly accessed by the CPU for program execution and also known as **Main memory** or **Primary Storage** and it is further divided into two parts:

- **RAM (Random Access Memory):** OS and programs in execution are kept temporarily in RAM and also known as **Volatile memory.**
- **ROM (Read Only memory):** It is a special type of memory and its content can't be changed, and it mainly contains manufacturer's instruction and 'Bootstrap Loader' ROM is also known as **Non-volatile memory**.

b) Secondary memory: It is a place in the computer where the data or programs are kept for long term and also a **Non-volatile memory**. It is of two types: Magnetic disks and Optical disks.

c) Cache memory: Cache memory acts as a buffer between CPU and main memory and stores the data that is frequently used so that the future request to that data can be served faster.

Cache Performance: Performance of a cache memory is measured by a term called Hit Ratio. If the processor finds the required memory location in the cache, then **Cache Hit** is occurred, and otherwise **Cache Miss** is occurred. In case of Cache Miss, a new entry is made in the cache and data is copied from the main memory to cache memory.

$$\text{HitRatio} = \frac{\text{Hit}}{\text{Hit} + \text{Miss}}$$

Input and Output (I/O) devices: I/O devices in a computer are used to receive raw information and data as input from the user and give processed and useful information as desired output to the users.

- **Input device:** Some of major examples of Input devices are **Keyboard** and **Mouse**. Some other example of input devices: Joystick, Track ball, Light pen, Scanner, Microphone, Bar code Reader, Optical Character Reader (**OCR**), Optical Mark Reader (**OMR**) and Magnetic Ink Character Reader (**MICR**).
- **Output device:** Some of major examples of Output devices are **Monitor** and **Printer**. Some other examples of output devices: Projector, Headphone, Plotter, Speaker, Sound Card and Facsimile (FAX)
- **Compute Software:** Set of program or instruction that enables computer to perform well defined functions and is classified into two categories: System Software and Application Software.
- **System Software:** System software is a group of programs that is designed to run a computer's hardware and provide a platform for other software. There are two major categories of System Software: **Operating System** such as Windows, DOS, Android and Apples iOS etc and **Utility Software** such as Device driver, Antivirus software, Disk defragmenter and Backup Software etc.

- **Programming Software:** Programming software is a program or set of programs which helps the software developers to develop other software. Programming Software are also known as **Programming tool** or **Software development tool**. Ex: Compilers, Interpreters, Debuggers, Text editors and Linkers etc.
- **Application Software:** Application Software is a program or group of programs that is designed for end users for a specific purpose. Some of the types of application software:

  Application Suite: An Application suite or Software suite is a group of different but interrelated software programs that are combined and packaged together. Ex: MS – Office

  Software in MS – Office Package: **MS – Word**, **MS – Excel**, **MS – OneNote**, **MS – Outlook**, **MS – PowerPoint**, **MS – Access**, **MS – Publisher** and, **SharePoint**.

i) Horizontal Software: Software that is widely used from individual users to big organizations and also known as **Productivity software**. Ex: MS – Office, Web browsers and Financial software etc.

ii) Vertical Software: Software that is designed to keep specific users in mind and targets a smaller number of users. Ex: ERP (Enterprise Resource Planning) and CRM (Customer Relationship Management) etc.

Software Licensing: Software licensing is legal instrument between the software application author and end user that governs the use or redistribution of software. It is also known as **End User License Agreement** (EULA).

Type of Software License:

- **Registerware:** Requires user's personal information in order to download or use the program through registration. Ex: Home edition of Avast.
- **Shareware: C**omputer software that is initially distributed on a free trial basis without payment with limited. After the trial period the user is obliged to pay fee to continue the use the software. Ex: Antivirus and WinZip etc. Type of Shareware: Adware, Donationware, Nagware, Freemium, Demoware (Cripple ware and Trialware).
- **Freeware:** Freeware is type of software that is distributed with no monetary cost small fee. Ex: Adobe reader and Skype etc.
- **Open Source Software:** It is a software that is distributed with its source code that can be modify and share by the users. Ex: Apache HTTP server, Mozilla Firefox and Blender etc.
- **Abandonware:** It is a software that is no longer sold or used by its owner and for which no product support is available from the manufacturer. Ex: Alley Cat and Old versions of photoshop etc.

License Key and Software piracy: Software license key is a data string that verifies and validates the authorization of software product purchase and Software piracy is the illegal or unauthorized use, copying and distribution of a software. Various type of software piracy is: Counterfeiting, Internet Piracy, End user Piracy, Client-Server Overuse and Hard-disk loading.

Computer Architecture referrers to those attributes of a computer system that are visible to the programmer such as instruction set, instruction opcodes, addressing technique, register and data memory etc and **Computer Organization** is concerned with the way operational units (Hardware) operate and their interconnection that forms the computer system. Computer Architecture tells "**What does a computer system do**" while the Computer Organization tells "**How to implement**".

Computer System Architecture: Based on the general-purpose processors used, types of computer system are of two types: Single Processor System and Multiprocessor System**.**

Multiprocessor systems are also known as **Parallel Systems** or **Tightly Coupled System**. There are two types of multiprocessor system: **Symmetric Multiprocessor System:** (Two or more identical processors are connected to a single shared memory) and **Asymmetric Multiprocessor System** (Not all the processors connected with each other are treated equally).

Clustered System: Like Multiprocessor system, clustered system is also composed of more than one processor to accomplish the task, but they are different from parallel systems. They are composed of two or more individual systems coupled together.

General System Architecture: Parallel processing is a method of breaking a problem in small parts and executing those parts simultaneously on different processors. Parallel systems are more difficult to program as compared to single processors system because the architecture of parallel processors is varying, and all the processors must be synchronised and co-ordinated. General System Architecture is mainly classified into two types:

1) Flynn's classification of computers: Based on number of instruction and data items that can be processed simultaneously, the organization of computer system is given below:

Organization of computer system	Definition
SISD Single-instruction, Single-data	Single processor system capable of executing a single instruction, operating on a single data stream. Instructions are executed sequentially.
SIMD Single-instruction, Multiple-data	Multiprocessor system and capable of executing same instruction on all the processors but operating on different data streams.
MISD Multiple-instruction, Single-data	Multiprocessor system capable of executing different instruction on different processor but all of them are operating on same data stream.
MIMD Multiple-instruction, Multiple-data	Multiprocessor system capable of executing different instruction on different processor and all of them are operating on different data stream.

2) Store Program Control Concept: Stored program concepts refers to the storage of instruction computer memory and allow it to perform various tasks in sequence automatically by processor. Classification of Stored Program concepts:

a) Von-Neumann Model: The computer architecture proposed by Von-Neumann is called Von-Neumann architecture. Von-Neumann model consists of a control Unit, Arithmetic and Logic Unit, Input/Output devices and Registers. Von-Neuman computers uses single processor, one memory for both instruction and data and execute programs following the fetch-decode-execute cycle.

Components of Von-Neuman model: There are mainly following components of this model: Central Processing Unit (ALU and CU), Memory and Registers, I/O devices, Buses.

Register is a type of storage unit that is used to store data and instruction in execution. Various type of registers used: **AC** (Accumulator), **PC** (Program Counter), **CIR** (Current Instruction Register), **MDR** (Memory Data Register), **MAR** (Memory Address Register) and **IBR** (Instruction Buffer Register).

Buses: Data from one part of the computer system to another part is transferred by the means of Buses. Three types of buses are used: Data Bus, Address Bus and Control Bus.

b) General Purpose System: General purpose system are the architecture of modern-day computer system and it is the modified version of Von-Neumann architecture.

Components of general-purpose system are Central Processing Unit (CPU) that consists Arithmetic/Logic Unit (ALU), Control Unit (CU) and various process register. All the components of general-purpose system are connected by the system busses (Data bus, Address bus and Control bus).

Computer Instruction: A processor performs tasks based on instructions provided and these instructions are a set of machine language instructions that a processor or computer can understand easily.

A basic computer has 16-bit instruction registers (IR) and an instruction can be divided into three parts:

- First 12-bits specifies the Address filed that stores the address of the memory where the data (operand) is stored.
- Next 3-bits specifies the Operation code (Opcode) that specifies the operation to be performed.
- Left most bit specifies the Mode field that specifies the addressing mode.
- I = 0 for Direct addressing (Address field stores the address of the operand.)
- I = 1 for Indirect addressing (Address field stores the address where the address of the operand is stored.)

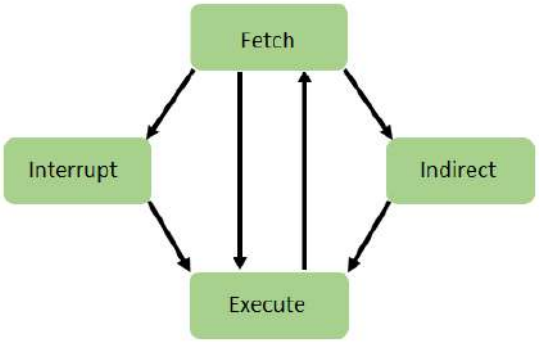
Basic computer has three type of instruction code format:

1) Memory reference instruction: In memory reference instruction, Address filed is denoted by 3 x's in hexadecimal notation which is equivalent to 12-bit address. These instructions refer to memory address as operand and another operand is always accumulator. Some of Memory reference instruction are: AND, ADD, LDA, STA, BUN, BSA, ISZ

2) Register reference instruction: In register reference instruction, first 12-bit represents the Register operations, next 3-bits represents the opcode which is equivalent to '111' and Left most bit represents the addressing mode which is equivalent to '0'. Last four bits (from left) instruction is always equal to hexadecimal 7. Some of Register reference instruction are: CLA, CLE, CMA, CIR, CIL, INC, SPA, SNA, SZA, SE, HLT

3) Input-Output instruction: In Input-Output instruction, first 12-bit represents the I/O operations, next 3-bits represents the opcode which is equivalent to '111' and left most bit represents the addressing mode which is equivalent to '1'. Some of Register reference instruction are: INP, OUT, SKI, SKO, ION, IOF

Instruction Cycle: Instruction cycle also known as fetch-execute cycle. A program in the main memory is a sequence of instruction and these instructions are executed by CPU by following the given instruction cycle and each instruction cycle consists of fetch, decode and execute phases.



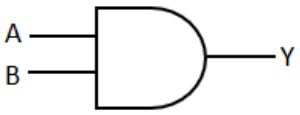
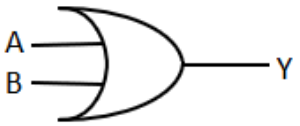
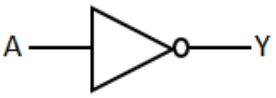
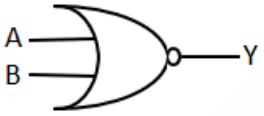
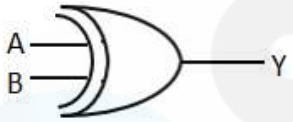
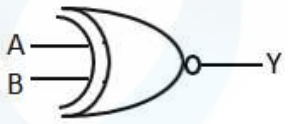
Digital System: Digital system is a system that operates on discrete levels of input represented in binary form- 0/1 or TRUE/FALSE or ON/OFF.

Boolean Algebra: Boolean algebra is a branch of algebra that deals with binary numbers and has truth values either TRUE or FALSE represented by '1' or '0' respectively and also known as Switching Algebra.

Boolean Laws: Boolean Algebra uses a set of laws and rules to define the operation of a digital logic circuit. The basic laws of Boolean Algebra are given below:

Boolean Laws	**Description**
Annulment Law	A.0 = 0 and A + 1 = 1
Commutative Law	A.B = B.A and A + B = B + A
Identity Law	A.1 = A and A + 0 = A
Associative Law	A.(B.C) = (A.B).C and A + (B + C) = (A + B) + C
Distributive Law	A.(B + C) = (A.B) + (A.C) and A + (B.C) = (A + B).(A + C)
Idempotent Law	A.A = A and A + A = A
Complement Law	A.A` = 0 and A + A` = 1
Double Negation Law	(A`)` = A
Absorption Law	A.(A + B) = A and A + AB = A
De Morgan's Law	(A.B)` = A` + B` and (A + B)` = A`.B`

Logic Gates: A logic gate is basic building block of any digital circuit and it mainly has two inputs and one output. There are three types of logic gate: Basic logic gates (AND, OR and NOT), Universal gate (NAND and NOR) and Special gates (XOR and XNOR).

Gate	Representation	Operation	Truth Table
AND		Y = A.B	**A** **B** **Y** 0 0 0 1 0 0 0 1 0 1 1 1
OR		Y = A + B	**A** **B** **Y** 0 0 0 1 0 1 0 1 1 1 1 1
NOT		Y = A`	**A** **Y** 0 1 1 0
NAND		Y = (A.B)` = A` + B`	**A** **B** **Y** 0 0 1 1 0 1 0 1 1 1 1 0
NOR		Y = (A + B)` = A`.B`	**A** **B** **Y** 0 0 1 1 0 0 0 1 0 1 1 0
XOR		Y = A ⊕ B = A.B` + A`.B	**A** **B** **Y** 0 0 0 1 0 1 0 1 1 1 1 0
XNOR		Y = A ⊙ B = A.B + A`.B`	**A** **B** **Y** 0 0 1 1 0 0 0 1 0 1 1 1

Number System: Number system is the technique of writing or expressing numerals of different types. There are various categories of numbers:

- **Binary numbers:** Binary numbers are expressed in **base 2** or **radix 2** numeral system. **Ex:** 10110_2 is a binary number.
- **Octal numbers:** Octal numbers are expressed in **base 8** or **radix 8** numeral system. **Ex:** 945_8 is an octal number.
- **Decimal numbers:** Decimal numbers are expressed in **base 10** or **radix 10** numeral system. **Ex:** 683_{10} is a decimal number.

- **Hexadecimal numbers:** Hexadecimal numbers are expressed in **base 16** or **radix 16** numeral system. **Ex:** $3D6A_{16}$ is a hexadecimal number.

Number System Conversion: The process of changing the base or radix of numbers is known as number system conversion:

1. Decimal to any other base system: Conversion process is given below:

- Divide the decimal number by new base value in which the decimal number is to be converted and record the remainder which will be LSB (least significant bit) of new base number.
- Continue to divide the quotient until the quotient becomes zero.
- Record the remainder of each division process from last to first and remainder of last division process will be MSB (most significant bit) of new base number.

Ex: Decimal to Binary: Convert 25_{10} to binary system-

2	25	
2	12	1
2	6	0
2	3	0
2	1	1
	0	1

$25_{10} = 11001_2$

2. Any other base to decimal system: Conversion process is given below:

- Determine the positional value of each digits according to the base of number system.
- Multiply the positional value by digits at the corresponding position.
- Add the product calculated in previous step.

Ex: Binary to Decimal: Convert 10110_2 into decimal-

$10110_2 = (1 * 2^4) + (0 * 2^3) + (1 * 2^2) + (1 * 2^1) + (0 * 2^0) = 16 + 0 + 4 + 2 + 0 = 22_{10}$

3. Binary system to Octal system: Conversion process is given below:

- Divide the binary digits into group of three from LSB (rightmost side).
- Write octal equivalent of each group to find the octal number.

Ex: Convert 1101011_2 into octal-

$1101011_2 = (1)\ (101)\ (011) = (001)\ (101)\ (011) = 153_8$

4. Octal system to Binary system: Conversion process is given below:

- Convert each digit of octal number to three digits binary equivalent.
- Combine all the groups of binary numbers into a single group to find the required binary number.

Ex: Convert 724_8 into binary-

$724_8 = (111)\ (010)\ (100) = 111010100_2$

5. Binary system to Hexadecimal system: Conversion process is given below:

- Divide the binary digits into group of four from LSB (rightmost side).
- Write hexadecimal equivalent of each group to find the hexadecimal number.

Ex: Convert 1101110_2 into hexadecimal-

$1101110_2 = (110)\ (1110) = (0110)\ (1110) = 6E_{16}$

6. Hexadecimal system to Binary system: Conversion process is given below:

- Convert each digit of hexadecimal number to four digits binary equivalent.
- Combine all the groups of binary numbers into a single group to find the required binary number.

Ex: Convert $B9_{16}$ into binary-

$B9_{16} = (1011)\ (1001) = 10111001_2$

7. Octal system to Hexadecimal system: Conversion process is given below:

- Convert each digit of octal number to three digits binary equivalent.
- Divide the obtained binary equivalent into the group of four from LSB.
- Write hexadecimal equivalent of each group to find the hexadecimal number.

Ex: Convert 65_8 into hexadecimal-

65_8 = (110) (101) = 110101 = (11) (0101) = (0011) (0101) = 35_{16}

8. Hexadecimal system to Octal system: Conversion process is given below:

- Convert each digit of hexadecimal number to four digits binary equivalent.
- Divide the obtained binary equivalent into the group of three from LSB.
- Write octal equivalent of each group to find the octal number.

Ex: Convert EC_{16} into octal-

EC_{16} = (1110) (1100) = 11101100_2 = (11) (101) (100) = (011) (101) (100) = 354_8

1. Weighted Binary code: In weighted binary code, 'weight' is assigned to each binary bit and value depends on the position of binary bit. **Ex:** Binary weighted code for Decimal 1 in 84-2-1 format = 0111 since 1 = 4 – 2 – 1. Hence, bits for '4', '-2' and '-1' will be on.

Decimal	8421	2421	84-2-1
0	0000	0000	0000
1	0001	0001	0111
2	0010	0010	0110
3	0011	0011	0101
4	0100	0100	0100
5	0101	1011	1011
6	0110	1100	1010
7	0111	1101	1001
8	1000	1110	1000
9	1001	1111	1111

Binary Coded Decimal (BCD) code: BCD is a type of binary representation for decimal value where each digit is represented by group of four binary bits.

Decimal	BCD	Decimal	BCD
0	0000	8	1000
1	0001	9	1001
2	0010	10	0001 0000
3	0011	11	0001 0001
4	0100	12	0001 0010
5	0101	13	0001 0011
6	0110	14	0001 0100
7	0111	15	0001 0101

2. Non-weighted Binary code: In Non-weighted binary codes, the positional weights are not assigned. Ex: Excess-3 (XS-3) code and Gray code.

Excess-3 Code: Excess- (XS-3) code is unweighted binary code that is obtained after adding $(0011)_2$ to binary of each digit or 3_{10} to decimal of each digit of a number. Binary form refers to the 8421 BCD of a number.

Excess-3 = (BCD + 0011) or (Decimal + 3)

Gray Code: Gray code is a binary code in which code for each successive pair is differed by one bit only. Hence, it is also known as **Unit distance code** or **Reflective code**.

Process of converting BCD to Gray code:

Let B = $b_3b_2b_1b_0$ is a binary number and G = $g_3g_2g_1g_0$ is Gray code, then-

- $g_3 = b_3$
- $g_2 = b_3 \oplus b_2$
- $g_1 = b_2 \oplus b_1$
- $g_0 = b_1 \oplus b_0$

Alphanumeric codes: These are binary codes also known as character codes used to represent alphanumeric data (alphabets, numbers and symbols). Commonly used alphanumeric codes are:

- **American Standard code of Information Interchange (ASCII):** ASCII code is used worldwide and is 7-bit long.
- **Extended Binary Coded Decimal Interchange Code (EBCDIC):** EBCDIC code is primarily used in large IBM computers and is 8-bit long.

1's complement: 1's complement of a numbers is derived after changing all 1's with 0's and all 0's with 1's.

Ex: 1's complement of '101101' is '010010'.

2's complement: 2's complement of a numbers is derived after adding '1' to the LSB of 1's complement of that number.

Ex: Calculate the 2's complement of '01010110'.

1's complement of 01010110 = 10101001

2's complement of 01010110 = 10101001 + 1 = 10101010

Binary arithmetic operations: Binary arithmetic operations are: Addition, Subtraction, Multiplication and Division.

Binary Operations	**Rules**
Binary Addition	

A	**B**	**Carry**	**Sum**
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Binary Subtraction

A	**B**	**Borrow**	**Difference**
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0

Binary Multiplication

A	**B**	**Multiplication**
0	0	0
0	1	0
1	0	0
1	1	1

Binary Division: Binary division is similar to the decimal division.

Boolean algebraic functions: A Boolean function is described by an algebraic expression consisting of binary variables. Boolean functions can be represented in two forms:

- **Canonical form:** Boolean expression in which each individual term contains every variable either in TRUE form or Complimented form is known as Canonical form. Ex: F = ABC + AB`C + A`BC`
- **Standard form:** Standard form is the simplified version of Canonical form. Ex: ABC + BC`

Two standard forms of Boolean function:

Standard Sum of Product (SOP): In the SOP form of Boolean function representation, variables are first operated by AND (product) gate and then these product terms are operated by OR (sum) gate, hence known as Sum of Product form. SOP form is also known as Disjunctive Normal Form (DNF) and logical OR operator in Boolean algebra is known as Disjunctive operator. **Example:** F = ABC + AB`C + AB + A`B`C`

Min-term: Each individual term in SOP form that contains every variable either in TRUE or Complimented form is called min-term. Sum of Product form is also called Sum of Min-terms.

Standard Product of Sum (POS): In this form POS form of Boolean function representation, variables are first operated by OR (sum) gate and then these sum terms are operated by AND (product) gate, hence known as Product of Sum form. POS form is also known as Conjunctive Normal Form (CNF) and logical AND operator in Boolean algebra is known as Conjunctive operator. **Example:** F = (A + B + C).(A + B` + C).(A + B).(A` + B` + C`)

Max-term: Each individual term in POS form that contains every variable either in TRUE or Complimented form is called max-term. Product of Sum form is also called Product of Max-terms.

Convert standard SOP to Canonical SOP:

- F = A + BC`
- F = A.(B + B`).(C + C`) + (A + A`).BC` **[A + A` = 1]**
- F = (AB + AB`).(C + C`) + ABC` + A`BC`
- F = ABC + ABC` + AB`C + AB`C` + ABC` + A`BC` **[A + A = A]**
- F = ABC + ABC` + AB`C + AB`C` + A`BC`

Convert standard POS to Canonical POS:

- F = (A + B).(A + C`)
- F = (A + B + C.C`).(A + B.B` + C`) **[A.A` = 0]**
- F = (A + B + C).(A + B + C`).(A + B + C`).(A + B` + C`)
- F = (A + B + C).(A + B + C`).(A + B` + C`) **[A.A = A]**

K-Maps: K-map is a table of rows and column with all its entries as 1's and 0's in the form of SOP and POS representation. Total number of cells in a K-map of 'n' variable is 2^n where each cell represents a value which corresponds to 'n' variable truth table value. If in a K-map,

- m_i is the min-term of a Boolean function, then place '1' at i^{th} cell of K-map.
- M_i is the max term of a Boolean function, then place '0' at i^{th} cell of K-map.
- di is the don't care of a Boolean function, then place either 'd' or 'x' at i^{th} cell of K-map.

2 variable K-map: K-map for function of two variables is given below:

Cells in two variable K-map = 2^2 = 4

A \ B	0	1
0	00	01
1	10	11

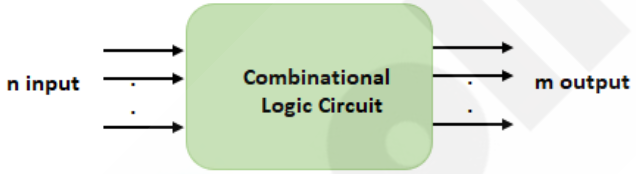
≡

A \ B	B`	B
A`	A`B`	A`B
A	AB`	AB

Similarly, we can maintain k-map for variables more than 2.

Combinational circuit: It is a circuit in which we combine different logic gates in the circuit and output of the circuit at any instance depends only on the present input given to the circuit irrespective of the previous input.

Block diagram of Combination circuit:



Binary Adder: In digital electronic, adder is a logic circuit that performs addition of binary numbers. There are two types of adder:

1) Half Adder: Half adder adds two binary input variables called **Augend** and **Addend** and produces two binary output as **Sum** and **Carry**.

S = A ⊕ B = A.B` + A`.B and C = A.B

2) Full Adder: Full adder takes three input variables; two input variables represent the two significant bits to be added and third bit is considered as the previous carry bit. It produces two output variables as Sum and Carry.

S = A`B`P + ABP + AB`P` + A`BP`

C = AB + AP + BP

Binary Subtractor: Binary subtractor is logic circuit used to perform subtraction operation on two binary digits. Binary subtractor can be designed using same approach as that of a Binary adder. There are two type of Binary subtractor:

Half Subtractor: It has two inputs; one is **Minuend** and other is **Subtrahend** and it produces two outputs; one is **Difference** and other is **Borrow**.

D = A ⊕ B = A.B` + A`.B

B_{out} = A`.B

Full Subtractor: It is logic circuit used to perform subtraction on three input bits; Minuend, Subtrahend and Borrow in (B_{in}). It produces two outputs; Difference and Borrow out (B_{out}).

$D = A \oplus B \oplus B_{in} = A`B`B_{in} + ABB_{in} + AB`B_{in}` + A`BB_{in}`$

$B_{out} = A`B + (A` \oplus B)\ B_{in} = A`B + A`B_{in} + BB_{in}$

Digital Comparator: It is combinational logic circuits that is used to compare two binary numbers and find the relationship between those two numbers.

1-bit digital comparator: Comparator used to compare two bits is called a single bit comparator.

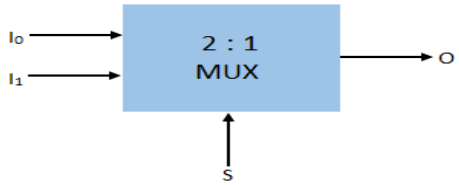
Truth Table:

A	B	C (A < B)	D (A = B)	E (A > B)
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

Multiplexer: A multiplexer is logic circuit that selects any one of the inputs given to circuit and passed it as output. Hence, it is also known as **Data selector** and selection of input depends on set of **Control signals** passed to the circuit which is also known as **Select inputs**. Multiplexer is also called MUX, Data selector, Parallel to serial convertor and many to one circuit.

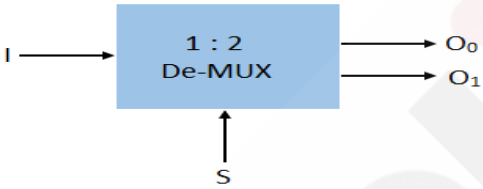
For 2^n input lines, there will be 'n' select lines and only one output.

Block diagram of 2: 1 MUX:



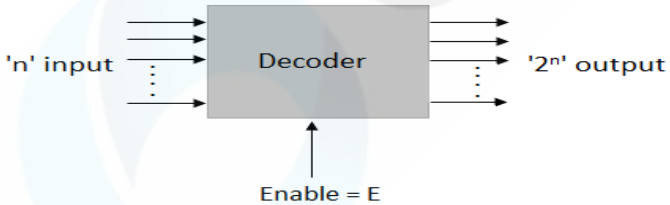
Demultiplexer: Demultiplexer is a combinational circuit that works opposite operation as that of Multiplexer. It has one input lines, 'n' selection lines and maximum of 2^n outputs. Demultiplexer is also known as De-MUX.

Block diagram of 1: 2 De-MUX:



Decoder: Decoder is combinational logic circuit that translate or decode a coded information from one format to another format. It takes 'n' inputs and can produce up to '2^n' coded outputs.

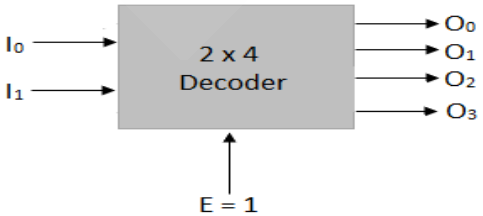
Block diagram of a decoder:



Decoder will work only when the value of E = 1 (Circuit is enabled).

2 to 4 Decoder: It is a decoder that takes 2 input and produces 4 output.

Block diagram:

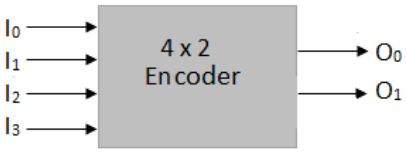


Encoder: Encoders is a combinational logic circuits that performs the reverse operation as that of Decoder. It has maximum of 2^n input and 'n' output lines and enable line is optional in the encoders.

Block diagram of Encoders:

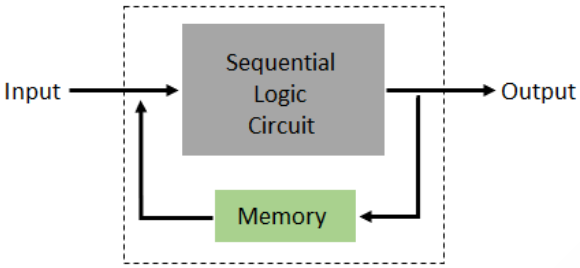
4 to 2 Encoder: It is a type of encoder that takes 4 input and produces 2 output lines.

Block diagram:



Sequential Logic Circuits: Sequential circuits are nothing but combinational logic circuit with memory. Memory is used to store the present state (Past Output) of the circuit and output of the circuit depends on the present input and present state (past output). The present output of the circuit is stored in the memory and works as next state of the circuit.

Block diagram of Sequential Circuits is given below:



There are two types of sequential circuits:

1) Asynchronous Circuit: Asynchronous sequential circuit runs with different clock pulse that means whatever sequential are there those elements are not going to use the same clock pulse. Ex: Latches.

2) Synchronous Circuit: Synchronous sequential circuit runs with same clock pulse or similar clock pulse. Ex: Flip-Flop.

Pipelining: Pipelining is a process where a program is divided into sub-programs (set of data processing elements) and then executed sequentially (serially).

Microprocessor: Microprocessor is an electronic device that incorporates the function of CPU on a single IC on a computer system and that computer system is known as **Microcomputer**. A microprocessor has three components: Arithmetic and Logic unit, Register Array and Control Unit.

Instruction set: Instruction in a microprocessor is a command intended to tell microprocessor to perform operations on given data. Instruction set is a collection of such instruction or commands in microprocessor.

Word length: Word length of microprocessor is a maximum number of bits it can accept as input, process or produce as output. Based on word length, processors are of two types: **RISC** (Reduced Instruction Set Computer) and **CISC** (Complex Instruction Set Computer)

Microprocessor Architecture: 8085 is widely known 8-bit microprocessor, designed by Intel in 1977 by using NMOS technology.

Configuration of 8085 microprocessor:

- 8-bit data bus and 16-bit address bus.
- 16-bit stack point and 16-bit program counter.
- Six 8-bit registers pairs BC, DE and HL.
- Require +5V direct current supply for its operation.

Architecture of 8085 microprocessor:

1. Arithmetic and Logic Unit (ALU): ALU consists of accumulator, flag register and temporary register. It takes input from memory and accumulator and then perform arithmetic operations such as addition and subtraction and logical operation AND, OR on them.

- **Accumulator:** It is denoted as 'A' and it is an 8-bit register used to stored 8-bit data.
- **Flag register:** Flag register is 8-bit register but only 5-bits are active.
    - **Sign flag:** S = 0 indicates the result is positive and vice versa.
    - **Zero flag:** Z = 1 indicates the result is zero and vice versa.

- **Auxiliary carry flag:** AC = 1 indicates the carry is generated and vice versa and it is used in BCD (0-9) number system.
- **Parity flag:** P = 1 indicates result is even parity and vice versa.
- **Carry flag:** CY = 1 indicates carry or borrow is generated and vice versa.

Interrupt Control: INTR, INTA, RST 5.5, RST 6.5, RST 7.5 and TRAP.

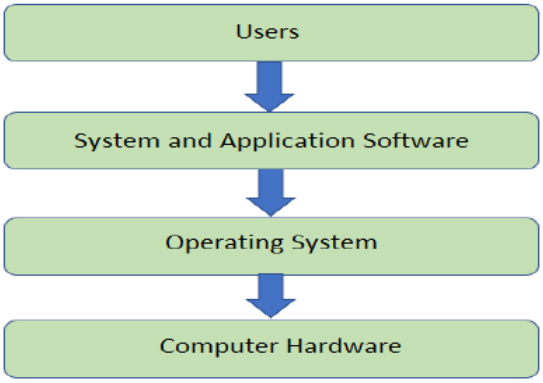
Serial I/O control: (Serial Input Data) and SOD (Serial Output Data).

Timing and Control unit: Generates timing and control signals and controls the flow of the data between microprocessor and peripherals.

Instruction register and decoder: Part of ALU and is not programmable.

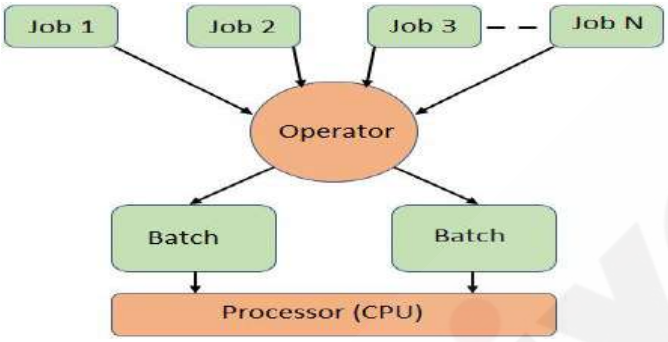
Lesson 2: Operating System

What is Operating System: An Operating System (OS) is a collection of software that manages software and hardware resources and provides services to computer programs and manages the resources (CPU, Main memory, Secondary storage and I/O devices) of computer. It is a software that acts as an interface between computer hardware and a user.



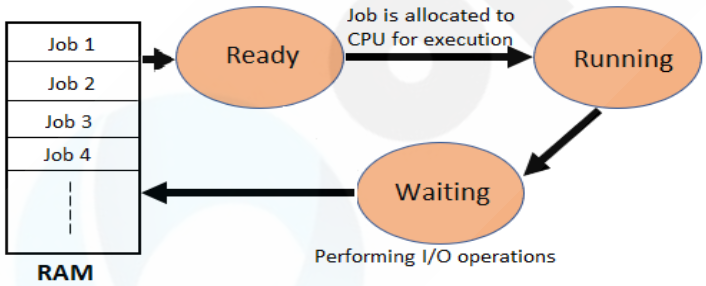
Type of Operating System OS: Some of the widely used operating System types are mentioned below:

1. Batch processing operating System: In Batch OS, user doesn't interact with OS directly. Users submit their jobs and an operator groups these jobs into batches. Jobs with similar requirements are grouped in the same batch and then these batches are processed by the processor one by one.

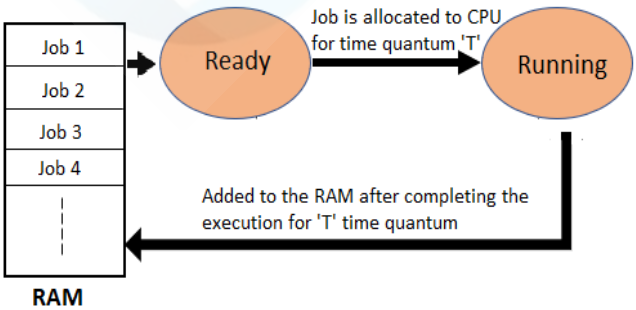


2. Multi-programming operating System: In this OS, more than one process can be processed by the processor simultaneously. The **Scheduler** decides which process to run by the processor and this process is known as **CPU scheduling**. The process of switching from one job to other job is known as **Context Switching**.

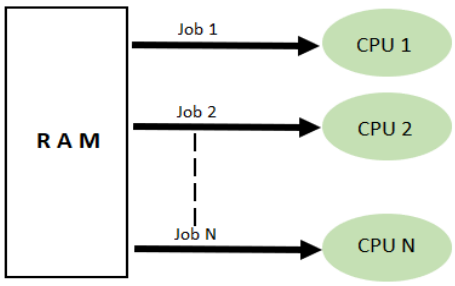
*** The main focus is to maximize the processor (CPU) usage.**



3. Multi-tasking operating System: Multi-tasking also known as **Time Sharing Operating system** is a logical extension of Multi-programming operating system. It allows users to share the resources simultaneously and in this, context switching is performed along with time sharing. **The main focus is to minimize the response time.**

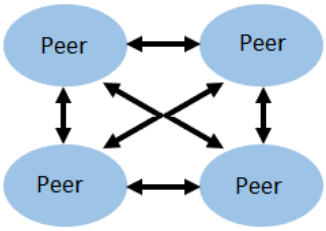


4. Multi-processing operating System: In multi-processing OS, more than one processor (CPUs) can be used within a single computer system.

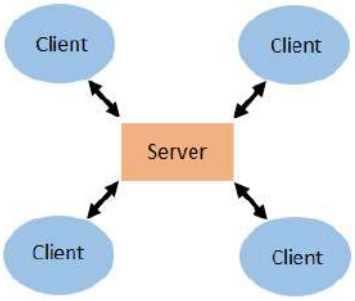


5. Network operating System: Network OS allows users to share resources available on their computer to other users and access resources available on other's computer.

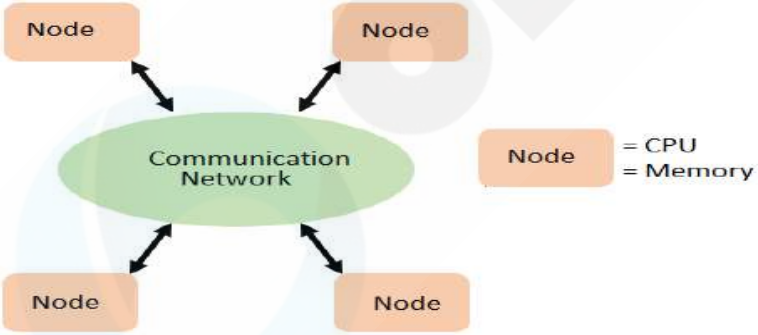
a) Peer – to – peer: In peer – to – peer, all workstations are considered equal and have same privilege to use the resources available on the network.



b) Client/server: In Client/server operating system, all functions and applications are centralized to single dedicated system called **Server**. All other systems (known as **Clients**) are connected to the dedicated system.



6. Distributed operating System: In Distributed operating system, various independent systems are connected with each other using a shared communication network but physically separated where they have their own processor and memory. A system can always access the resources that are not available on their system but on other system in the network. This type of OS is an advancement of Network operating system and also known as **Loosely Coupled System**.



7. Real Time operating System: Real time operating system (RTOS) are used in applications where the result is guaranteed in specified time constraint. These are very fast and quick in response and used in applications where large events are to be executed in a short time.

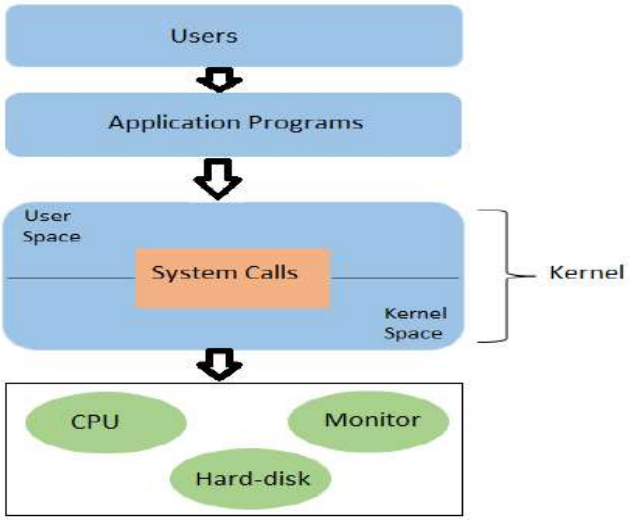
- **Hard real time OS:** In Hard real time OS, the process should be executed within specified time and it can't miss its deadline. Missing the deadline can cause disasters consequences and system will fail. Ex: Anti – lock brakes, Flight control system.
- **Soft real time OS:** In Soft real time OS, it is not mandatory that an execution must be finished in the specified time every time. Ex: Digital camera, telephone switches.

Components of OS: All the user software programs can't directly access the computer hardware; they need to go through OS to access the hardware. Various components of operating system are:

1. Kernel: Kernel is a central component in the computer's OS. It is a computer program that has complete control over everything of computer system. It is the first program after boot loader that loaded, and its critical code is

loaded in the protected area of the memory called **Kernel Space** while other application programs are loaded in separate area of memory called **User space**.

Kernel is a low-level abstraction layer of OS, user uses **System calls** to interact with the system, system call invokes the kernel and then kernel perform the user's operation.



Based on their design and functions, Kernel is divided mainly into three parts:

- **Monolithic Kernel:** A monolithic kernel performs all the OS instructions in the same address space and hence increases the speed.
- **Micro Kernel:** A microkernel runs mostly all the processes in User space and hence increases the Modularity.
- **Hybrid Kernel:** A hybrid kernel is an extension of Micro kernel with some added functionality of Monolithic kernel.

2. Interrupts: An interrupt is an input signal that indicates an event and need the immediate attention of the processor. When an interrupt occurs and request is accepted, the processor halt the current execution immediately and executes a function called **Interrupt handler**. After the interrupt handler finishes its activities, the processor resumes its normal activities.

There are mainly two type of Interrupts in operating system:

- **Software interrupts:** Caused by software itself which is usually a program code in the user space. Software interrupts are typically called **Traps** or **Exceptions**.
- **Hardware interrupts:** An electronic signal created and sent to the processor by performing some actions by the hardware components are known as Hardware interrupts.

3. Program Execution: OS provides an interface between application software and hardware components. Program execution is the process by which a computer executes the instructions of a computer program.

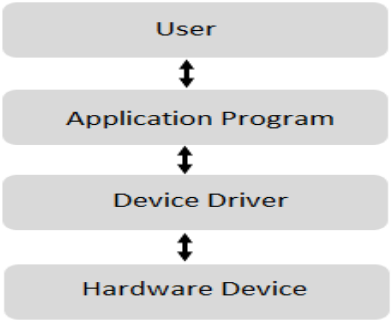
4. Memory management: It is the process of controlling and handling the primary memory. It is the process of partitioning the memory into small blocks and assign these blocks to a running process to optimize the overall system performance.

All the programs must be loaded into the main memory (RAM) for its execution. Sometimes the whole program is loaded into the RAM while sometimes a part of the program is loaded when called by the program and this process is known as **Dynamic Loading**.

Sometimes a running program need another program that is not present in the main memory. So instead, of loading all the related programs in the main memory, processor links the related programs to the executing program in the main memory, this process is known as **Dynamic Linking**.

5. Disk access and file management: The systematic way in which the data is stored in available storage space is called **File system**. File system is the process in which the files are named properly and stored logically in the disk for easy storage and fast retrieval.

6. Device drivers: Hardware devices need a program to communicate with the OS of the computer system. A computer program with which the hardware devices such as printer can communicate with the kernel of computer system are known as **Driver** or **Device driver**.



The device drivers can be broadly categorized into three groups:

- **Character driver:** Deals with the character devices that communicate by sending and receiving single characters.
- **Block driver:** Deals with the devices in that communicates by sending and receiving entire block of data.
- **Network driver:** The network driver allows us to connect to an external network by sending and receiving data packets.

7. User interface: A user interface is that part of operating system by which a user interacts with the computer system and provides platform for Human – computer interaction. There are different types of user interfaces:

- **Command line interface (CLI) or Character user interface (CUI):** It is the oldest user interface, which is a text-based user interface which involves the commands types by the user and respond of computer based on those commands. Ex: Disk operating system (DOS).
- **Graphical user interface (GUI):** GUI is the most attractive way of interaction between user and computer. It deals with the graphical images, icons and it allows users to interact with the machine by those images, icons and visual indicators. Ex: Windows and Android.
- **Menu driven interface (MDI):** A menu driven interface is the simplest and easiest way to communicate with the computer system. It deals with the series of screens (menus) where user has to choose one option what to do next out of available options in the menu. Ex: Cash machines (ATM), ticket booking machine and information kiosk.

Functions of Operating System: Following are some functions of operating system:

1. **Booting:** Booting is the start-up sequence of operating system and boot process loads the operating system into main memory of the computer system.
2. **Loading and execution of program:** A program resides in the secondary memory must be loaded in the main memory of computer for execution.
3. **Security:** Security refers to integrity, confidentiality and availability of data which of provided by Operating system.
4. **Control over system performance:** Operating system has complete control over computer system so as to improve the performance of the system.
5. **Job accounting:** The operating system keeps track of time and resources used by a process and maintain this information in a log record.
6. **Acts as an interface between software and user:** It coordinate between software and user and assigns software such as assembler, interpreter, compiler and other software whenever user need them.

Some other major important functions of Operating System:

- Process management
- Memory management
- File management
- I/O device management

Program: A computer program is lines of codes or collection of instructions that performs a specific task when executed. A process is a program in execution, any program written by programmer reside in the hard disk or secondary storage of a computer. A processor can execute programs in the main memory, so a program must bring into main memory from secondary storage.

A process in main memory can be divided into four sections:

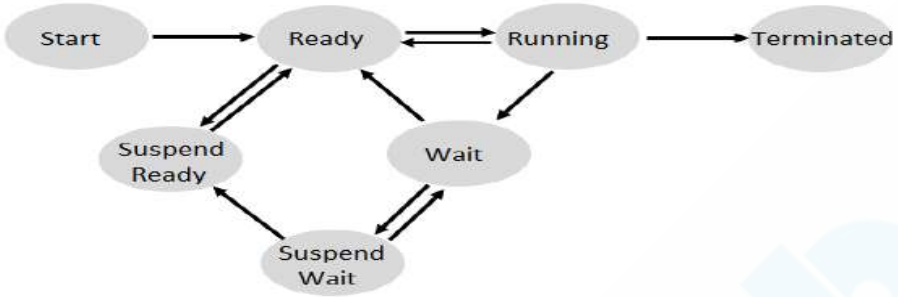
- **Stack:** A process stack contains temporary data such as functions/methods parameters, return value/address, local variables etc.
- **Heap:** Heap is an area of main memory that is assigned dynamically at the runtime of process to store temporary variables and data.

- **Data:** Data section of a process contains global and static variables.
- **Text:** Text section of a process includes the current activity represented by the value of Program counter and it also includes the content of Processor's register.

Process Control Block (PCB): In a multiprogramming OS, track of all the processes must be kept and PCB is a data structure maintained by OS of each process which contains all the attributed or characteristics of a process.

- **Process Id**
- **Process State**
- **Program Counter**
- **Priority**
- **Registers**
- **List of Open devices**
- **List of Open files**

Process Life cycle: A process in execution from its completion passes through different states in its life cycle:



Process Scheduling: Process scheduling is the act of process manager that determine the removal of a running process and inclusion of a new process into the main memory. A single processor can execute one process at a time and remaining process must wait until the CPU gets free. But in multiprogramming system, more than one processes can be executed simultaneously and hence to increase CPU utilization, there is a need of Process scheduling.

Type of Scheduling: The main perspective of the process scheduling is to keep the processor (CPU) busy all the time. There are two types of process scheduling:

- **Non-Pre-emptive Scheduling:** A process which is assigned CPU will complete its execution and another process in the ready queue will wait for its turn for its execution.
- **Pre-emptive Scheduling:** Processor is assigned to a process for a limited time and then it is moved back to the ready queue and process is assigned to some other process for limited time.

Queues used in Process scheduling: There are three type of queues used in process scheduling:

a) Job queue: All the processes enter into the system are kept in this queue.

b) Ready queue: All the processes that are in the ready state and waiting for processor for its execution are kept in this queue.

c) Device queue: The processes in the block/wait state waiting for I/O device are kept in this queue.

Process types: Based on the CPU utilization and I/O device utilization, processes can be divided into two types:

a) **CPU bound process:** Processes whose CPU utilization is more then I/O device utilization.

b) **I/O bound process:** Processes whose CPU utilization is less then I/O device utilization. These types of processes mainly reside in the block/wait state.

Process schedulers: The process schedulers are programs that schedules processes from one state to another state. There are three types of process schedulers used in process scheduling:

a) Long term (Job) scheduler: Job scheduler brings the newly created processes to the Ready state.

b) Short term (CPU) scheduler: CPU scheduler selects one process from the ready queue and allocates CPU to them for execution. Short term scheduler is also known as **Dispatcher.**

Context Switching: Process of storing the state of a process so that at a later time the process can be resumed from the same point which is done by Dispatcher.

c) Medium term scheduler (Swapper): Medium term scheduler removes the processes from main memory to secondary storage and vice versa and this process is known as **Process Swapping,** It is also known as **Swapper**.

Scheduling Criteria: The Scheduling criteria for process scheduling is mentioned below:

- **CPU utilization:** CPU utilization must be kept as high and it ranges from 0% to 100% and CPU must be kept busy all the time.

- **Throughput:** Number of processes that completes its execution per unit time.
- **Turnaround time:** Time taken by process to complete its execution from its submission.
- **Waiting time:** Time taken by a process in the ready queue from its creation waiting for allocation of process.
- **Response time:** Time taken by processor to response for the first time from the time when the request for the processor is made by the processor.
- **Load average:** Average number of processes waiting in the ready queue for their turn.

Scheduling Algorithms: There are various process scheduling algorithm based on two factors: Non pre-emptive CPU scheduling and Pre-emptive scheduling.

1. First come first serve (FCFS) CPU scheduling: In FCFS scheduling algorithm, the process which arrives first, get executed first. First come first serve follows FIFO queue data structure.

Note: Performance of scheduling algorithm depends on Average time, Lower the Average Waiting Time, better the scheduling algorithm.

2. Shortest Job First (SJF) scheduling algorithm: In FCFS scheduling algorithm, CPU is allocated to the processes according to their Arrival time but in Shortest Job First (SJF) scheduling algorithm, CPU is allocated to the processes according to their CPU burst time. Process with lowest CPU burst time will get the CPU first.

3. Shortest remaining time first (SRTF) scheduling: Shortest remaining time (SRT) is the pre-emptive version of SJF scheduling. The process with least remaining CPU burst time is executed first by the processor. At the arrival of each process, remaining CPU burst time of all the ready process is compared and processor is allocated to the process with least remaining CPU burst time.

4(a). Priority based Scheduling (Non-pre-emptive): In Priority scheduling, each process is given a priority and CPU is assigned to a process according to their priority. Process with high priority will get the processor first and so on. Priority of a process can be decided on the basis of memory requirement, time requirement or remaining CPU burst time. Processes with the same priority are executed on first come first serve basis.

4(b). Priority based Scheduling (Pre-emptive): In Pre-emptive priority scheduling, at the time of arrival of new process, priority of the running process is compared with the newly arrived process and priority of other waiting process in the ready queue. If the running process has highest priority, then it continues its execution otherwise it is taken out from the processor and other process with high priority gets the processor for execution.

5. Round Robin (RR) Scheduling: Round Robin scheduling is the pre-emptive version of first come first serve scheduling. Each process is provided a fix amount of time called time **quantum** for its execution. If a process finishes its execution in that time quantum, then the process terminates otherwise the process go back to the ready queue and wait for its next turn to complete its execution.

6. Highest Response Ratio Next (HRRN) scheduling: In this scheduling algorithm, processor is allocated to a process on the basis of **Response Ratio** of that process. A process with higher response ratio will get the processor first.

Response ratio = (Waiting time + CPU burst time)/CPU burst time

Thread: A thread is a path of execution within a process, it contains its own program counter, stack and set of registers. A thread is also known as lightweight process, a process can be divided into multiple threads to increase parallelism. The CPU switches rapidly between the threads of a process and gives us an illusion that the threads are running in parallel.

Type of threads: There are two type of threads:

- **User Level Thread:** User level threads are managed by user level library (applications) and its implementation is easy. If user level thread performs blocking operation, then the entire process will be blocked.
- **Kernel Level Thread:** Kernel level threads are managed by integral part of operating system (kernel) and its implementation is not easy. If kernel level thread performs blocking operation, then another thread can continue its execution.

Multithreading model: When multiple parts (threads) of a process are allowed to execute simultaneously, this is known as Multithreading model. There are three types of Multithreading models:

- **One to one:** In one to one model, each user level thread is mapped with a kernel level thread.
- **Many to one:** In many to one model, many user level threads are mapped with a kernel level thread.
- **Many to many:** In many to many, many user level threads are mapped with equal or lesser number of threads.

* Many to many doesn't have the disadvantages of one to one and many to one multithreading models.

Memory Management: CPU can process that are in the main memory while a program mainly resides in the secondary storage. For its execution it must be brought into the main memory hence, **Memory Management** is the

technique by which the OS manages the main memory of the system. Once the execution of process is completed by the CPU, the process is moved back to the secondary storage by OS

Logical and Physical addresses in operating system: Logical address is generated by the CPU while the Physical address represents a physical location in the memory.

Address Binding: Process of mapping logical address to corresponding physical address and address binding is done by three ways:

- **Compile Time:** Process of translation of logical address to physical address at the time of process compilation. If we know at compile time where the process will reside in main memory, then **absolute code** is generated, and address is translated at the compile time.
- **Load Time:** If it is not known where the process will reside in main memory, then binding is delayed till load time and compiler generates **relocatable code** and address is translated at the load time.
- **Execute/Run Time:** In multi-programming systems, there is need to shift physical address from one location to another location, then binding is delayed till run time or execute time.

Compile time and Load time address binding generates similar logical and physical address while Execute time address binding generates different logical and physical address. Set of all logical addresses and all physical addresses is called **Logical Address Space** and **Physical Address Space** respectively.

Loading and Linking: Carrying the executable program from secondary storage into main memory for execution is called Loading while Linking is the process of binding all the different modules of a process to a single executable module.

Swapping: Process of swapping out processes from the main memory and moved to the secondary storage and later the process is swapped back to the main memory from the secondary storage. Swapping is useful to run multiple program on a single processor simultaneously and is also called as the **Technique for Memory Compaction**.

Swap time is mainly the transfer time of processes between main memory and secondary storage and transfer time is directly proportional to the amount of memory swapped.

Contiguous Memory Allocation: In contiguous memory allocation, all the available memory space is not scattered and is available at one place and each process is contained in a single contiguous section of memory. Contiguous memory allocation is of two type:

1. Fixed/Static size partition: Simplest form of memory allocation, where memory block is divided into several fixed size partitions of same or different size and one partition can contain one process at a time. When the partition is available, a process from ready queue is loaded into that partition and once the execution is finished, the process is moved back to the secondary storage and partition becomes available for another process. The processes that are coming from ready queue must be of size equal to or less than the size of partition because **spanning** is not allowed in contiguous memory allocation.

Spanning: A part of process is allocated to one part and remaining part is allocated to some other partition.

Limitation of Fixed size partitioning:

- If size of a processes is less than the allocated block size of memory which leads to wastage of memory and known as **Internal Fragmentation**.
- Total size of scattered memory block is more than the new arriving process, but the memory can't be allocated since spanning is not allowed which leads to **External Fragmentation**.
- Fixed **Degree of multiprogramming.**
- Limitation on **Size of a process**.

Whenever there is internal fragmentation, external fragmentation will be always there.

2. Variable/Dynamic size partition: In this, partition is not made before the execution. Whenever the processes are coming, then the space is allocated to the processes. Partitions are made during the run time and according the size of process.

Advantage/Disadvantage of Fixed size partitioning:

- No Internal Fragmentation.
- No limitation on Degree of Multiprogramming.
- No limit on size of process.
- External Fragmentation.

Algorithms used in memory allocation: There are basically four algorithms of memory allocation: First fit, Nest fit, Best fit and Worst fit.

Non-Contiguous Memory Allocation: Free memory space is not available at one place and it is scattered in the main memory and spanning is allowed in non-contiguous memory allocation. Internal and external fragmentation is not occurred in non-contiguous memory allocation. Non-contiguous memory allocation is also known as **Virtual**

Memory allocation as it gives an illusion to the user that his whole program is loaded into the main memory which is not true.

Non-contiguous never means not contiguous. If contiguous memory blocks are available, then memory will be allocated contiguously.

Classification of non-contiguous memory allocation:

Paging: Paging is a fixed size partitioning scheme and in which a process is divided into sub-processes known as **Pages** before it is loaded into the main memory and memory is also divided into partitions of same size known as **Frames**. Size of each frame must be equal to the size of each page and now, each frame is allocated to each page of the process to avoid the external fragmentation and maintain the optimum use of the memory.

Logical address is mapped with physical address by **Memory Management Unit (MMU)** that uses **Page Table** for this purpose. Each process has its own page table that contains the frame numbers which tells where a particular page is present in the main memory.

Advantage/Disadvantage of Paging:

- It suffers from internal fragmentation because it has frames of fixes size.
- It reduces external fragmentation.
- Swapping is easy due to equal size of pages and frames.

Demand paging: In paging, only few pages are stored in the main memory while remaining are in the secondary storage and gives an illusion that main memory is big enough to store the complete program, this is called Virtual memory. **Demand paging** or **Virtual Memory Management** is a concept that decides which pages need to be kept in the main memory and which need to be kept in the secondary storage.

Page Fault: If the referred page is not present in the main memory, the **Page Fault** or **Page Miss** occurs, and that missing page is swapped from secondary storage to main memory.

Thrashing: When page fault is very high, CPU remains busy in swapping pages and effective access time will be equal to the time taken to swap pages which is called **Thrashing**.

Page Replacement Algorithm: Technique which decide which page need to be swapped out when a new page is loaded into main memory. If a referenced page is not found in the main memory, page fault occurs and that referenced page from the secondary storage is loaded into main memory that replaces one existing page in the main memory.

- **First In First Out (FIFO):** Operating system keeps track of all the pages in the main memory in a queue. Oldest page in the main memory is always selected for replacement.
- **Optimal Page Replacement:** In this algorithm, pages that would not be used for longest period of time are replaced.
- **Least Recent Used (LRU):** In this algorithm, the pages which in the main memory which are not used from the long period of time are selected for replacement.

Belady's Anomaly: In all page replacement algorithm, number of page faults decreased when number of frames are increased while in case of FIFO page replacement algorithm, number of page faults increases when number of frames are increases and this strange behaviour is shown by FIFO algorithm in some cases which is called **Belady's Anomaly**.

Segmentation: Memory management technique in which a process is divided into variable size part and each part is known as **Segment**. Each segment has its own segment table that contains two information: **Base address** and **Size** of segment.

In segmentation, each segment contains one complete module and when that module is loaded into main memory there is no need to load another segment and thus reduces page fault.

Advantage/Disadvantage of Segmentation:

- No internal fragmentation.
- Less page faults.

I/O Management: The role of OS in a computer's I/O management is to manage I/O operations and I/O devices. In computer system, I/O devices are responsible for accepting inputs from users and presenting output to the user based on the mentioned operations performed on the input. I/O devices can be divided into two types:

- **Block oriented devices:** Block devices performed the communication by sending entire block of data (chunk). **Ex:** All the disks should be considered as the block-oriented device.
- **Character oriented devices:** Character devices performed the communication by transmitting single byte of data. **Ex:** Mouse, Printer and Keyboard.

I/O Hardware: I/O hardware devices can be categorized into:

- Storage devices (tapes and disks)

- Transmission devices (modems and network cards)
- Human-interface devices (monitor, keyboard and mouse)

I/O hardware devices are connected with each other using PCI bus. An I/O port typically consist of four registers:

- **Data-in register:** Read by the host to get input from the devices.
- **Data-out register:** Written by the host to send output.
- **Status register:** It contains bits that is read by host and these bits indicates the states such as: ready, bust, idle, transaction complete and error etc.
- **Control register:** It contains bits that is written by the host to start command or to change the mode of the device.

The process of detecting the arrival of input can be done by two ways: Polling and Interrupts.

Pooling: Process in which the processors periodically checks for the status of devices to determine whether any I/O operation is requested or not. Processor uses **busy bit** in **status register** to indicate whether it is available to execute I/O operation or not. Busy bit 1 indicates processor is performing some other function and bust bit 0 indicates processor is ready to execute I/O operation.

Host devices uses **command-ready bit** in the **command register** to indicates whether the host is ready with it I/O data or not.

Interrupts: Interrupts are another method by which I/O request of the devices can be handled by the micro-controller (processor). The processor uses **interrupts-request line** to detect the request generated by I/O devices. When processor senses any interrupt request on the interrupts-request line, it performs save state to the current execution and jumps to the **interrupt-handler routine**. Performs the desired I/O operations and after servicing the interrupt, return to the execution state prior to the interrupt.

Polling and Interrupts are useful when relatively small amount of data needs to be transferred between CPU, main memory and I/O devices.

I/O Software: I/O software are classified into three layers:

- **User level libraries:** This provides an interface to the user program so that input and output operations can be performed.
- **Kernel level modules:** It contains device independent operating system software, Device drivers and Interrupt handlers.
- **Hardware:** At this layer, hardware and hardware controller exists which interact with the device driver and makes software alive.

Kernel I/O Subsystem: The kernel is a core of OS and provides many services to other part of OS and the kernel I/O subsystem is responsible for many services related to I/O as give below:

- **I/O scheduling:** Kernel receives various I/O request and arrange them in good order to increase the overall system efficiency. This I/O scheduling is also known as **Disk Scheduling**.
- **I/O protection:** Kernel I/O subsystem provided I/O protection from illegal I/O instructions and ensures that I/O must perform only on system calls.
- **Buffering:** Kernel I/O subsystem maintains a memory called **buffer** in between storage devices and data is stored in buffer while transferring between devices.
- **Caching:** Caching is a process of holding the copy of a data for faster access and for this a cache memory is placed that stores the frequently used data.
- **Error Handling:** An operating system can provide error handling for various failures such as disk read/write, buffer overload, device unavailable and Device and I/O transfer.
- **Spooling:** Spool is a buffer that holds the output of a process for devices that can't accept interleaved data and spooling process copies the entire queued spool once at a time to that device.

Disk Scheduling: It is also known as I/O scheduling and is one of the important functions of operating system. In I/O scheduling OS schedule the I/O request for the disks.

Important terms in Disk scheduling:

- **Seek Time:** Seek time is the time taken to look for the disk arm to a specified track where the data needs to be accessed (read or write).
- **Rotational Latency:** Rotational latency is the time taken by desired sector of the disk to rotate itself to the position where it can access the read/write head.
- **Transfer Time:** Transfer time is the time taken to transfer the data and it depends on the number of bytes to be transferred.
- **Disk Access Time:** Disk access time is the sum of Seek time, Rotational Latency and Transfer time and is give as:

Disk Access Time = Seek Time + Rotational Latency + Transfer Time

- **Disk Response Time:** Disk response time is the average of time spent by each request waiting for I/O operation. **Mean response time** is the response time of all the requests and **Variance response time** is the measure of how each request is serviced with respect to mean response time.

Disk Scheduling Algorithms: There are various disk algorithms given below:

- **First Come First Serve (FCFS) scheduling:** It is simplest scheduling algorithm and follow the FIFO policy, I/O requests are serviced in the order in which they arrive. As every request in the waiting queue is serviced so there is no starvation and every request are honoured.
- **Shortest Seek Time First (SSTF) algorithm:** SSTF is improved version of FCFS scheduling. In this, the seek time of each request is calculate before scheduling and then the requests are scheduled according to their seek time, request which requires least disk arm movement from its current position regardless of direction is scheduled first and so on and hence reduces the total seek time.

File Management: File is a collection of logically related information that is stored in the secondary storage. File management is the process of manipulating (creating, modifying and deleting) the files in the computer system. Files in the computer system of different types:

1. Ordinary File: Most common files used to store either text or binary data. Ordinary files are also known as **Regular files** or **Simple files**.

- **Text files:** Files that contains information in ASCII format and that can be read by the user.
- **Binary files:** Files that contains data or information in binary format and are readable by the computer system.

2. Directory File: Directory files are the files that contains information about files or folders in the system but doesn't contain the actual file data.

3. Device File: Device files are also known as **Special files** and these are the files which represents the physical devices such as disks and printers. Devices files are of two types:

- **Character special files:** These are the device files which read or write data character by character. Ex: Mouse and Printer.
- **Block special files:** In these type of special files, data is handled in blocks. Ex: Disks and Tapes

Attributed of file: A file has following attributes:

- Name
- Type
- Location
- Identifier: Identifier is also known as extension of file such '.txt', '.jpeg', '.exe' and '.mp3' etc.
- Size
- Protection
- Time and Date

File Structure: A file has various kind of structure as given below:

- **No define Structure:** It contains sequence of bytes and words.
- **Simple record Structure:** It contains lines of fixed and variable length records.
- **Complex Structure:** It contains formatted document and relocatable load files.

File Allocation Methods: OS allocates disk spaces to the file and there are various methods which are used to allocate disk space to any file which are given below:

- **Contiguous Allocation:** In this allocation method, block of memory is allocated to the file in such a way that all the logical blocks of the file get the contiguous physical block of the memory.
- **Linked Allocation:** In linked allocation, each file carries a list of links to disk blocks and the disk blocks allocated to a particular file need not be contiguous in the memory. The directory contains a pointer to the first block of the file which points to the next disk block allocated to the same file.
- **Indexed Allocation:** There is a problem with linked allocation that random access is not allowed in it since, each block can be found from the previous block. An index block is created that contains all the disk pointers to file. Index block holds the pointers to all the disk block allocated to that particular file.

Operating System Virtualization: Virtualization in operating system allows to run different type of applications on a single computer system by many users and operating system may appear different to each user.

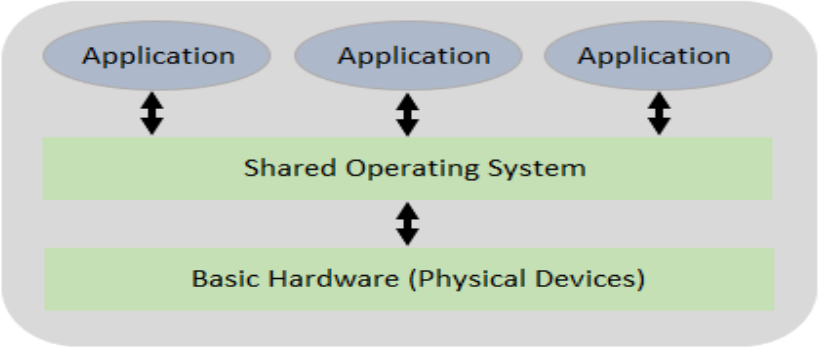


Fig: Operating System Virtualization

Operating System Security: Security management is a function to make computer system secure and protect internally as well as externally. Security refers to provide protection to computer resources such as CPU, memory, software and hardware program and most importantly the data stored in the system against unauthorised access and malicious programs.

1. Authentication: Authentication means the person using the computer system is genuine and allowed to use that computer system. This can be done in different ways: Username/Password, One Time Password (OTP) and Biometric data.

2. Access Control: Access control decides who or what can access (view or use) the resources of computer system.

3. Program and Software Threats: When user program designed to perform malicious task, it is called program threat. While system threat referred to misuse of system services that leads to loss or modification of data or physical damage to any hardware device.

a) Type of Program threats:

- **Virus:** Virus is any program inserted in user's computer without his/her knowledge and performs malicious functions when executed. It can replicate itself and can modify other computer programs.
- **Trojan Horse:** Malware that presents itself as a legitimate program and tricks user to enter his/her confidential details which later can be used for malicious activities.
- **Logic Bomb:** Logic bomb is a piece of code that is inserted into user's computer system and execute itself when certain condition is met which often lead to harmful effects.
- **Trap Door:** Trap door is also known as Back Door which provides a secret and unauthorised access to unauthorised user without the knowledge of authorised user.

b) Type of Software threats:

- **Denial of Service (DoS):** In DoS attack attacker makes the resources unavailable to its legitimate user.
- **Port Scanning:** Port is a place (door) where information goes in and comes out from computer system and port scanning is a process to scan all the ports to check for open doors to break into the computer system.
- **Worm:** Worm is a standalone malware computer program that replicate itself and uses computer network to spread itself to other computers connected to that network.

Type of Security breaches:

- **Breach of Availability:** Unauthorised access of data and resources and its destruction.
- **Breach of Confidentiality:** Theft of private or confidential information/data.
- **Breach of Integrity:** Unauthorised modification of private or confidential data.
- **Denial of Service:** Prevents the legitimate use of the information to the authorised user.
- **Theft of Service:** Unauthorised use of the information by an unauthorised user.

Process Synchronisation: Process Synchronisation is a way in which concurrent access to resources by the processes are handled and shared data consistency is maintained. Data inconsistency can result to Race Condition.

Race condition: Two or more processes try to access the same resource without any proper sequence.

Critical Section: A critical section is a code segment that can be accessed by only one process at a certain instance of time.

Critical Section Problem: It contains the shared resources that is to be accessed by the other processes as well and execution of shared variables need to be synchronised so that only process at a time can enter into the critical section and other processes are allowed to enter the critical section only after when previous process releases the shared resources.

```
do
{
   enter section;
      critical section;
   exit section;
      remainder section;
}
while (true);
```

Solution to Critical Section Problem: Any solution to critical section problem must follow the following three conditions:

- **Mutual Exclusion:** From a group of processes only one process at a time can enter into critical section and no other process is allowed to execute in the critical section.
- **Progress:** If no process is executing in the critical section, then only those waiting processes that are not executing their remainder section can participate in the selection of process that can execute its critical section and selection can't be postponed indefinitely.
- **Bounded Waiting:** Bounded waiting means no process should wait for the allocation of resource for indefinite amount of time. There must be bound on the number of time the process can enter into critical section.

Semaphores: A semaphore S is an integer variable that can be accessed through only two operations wait() and signal(). The wait() operation is termed as P(S) and signal operation is termed as V(S).

Definition of wait() and signal() variables is as follows:

```
wait(S)
{
   while S <= 0
      no operation;
      S--;
}
signal(S)
{
      S++;
}
```

Bounded Buffer Problem: It is also known **Producer-Consumer problem** in which there are two processes: Producer and Consumer and both share a common fixed size buffer. Producer tries to put data into empty slot of buffer and consumer tries to remove data from filled slot of buffer. The problem is Producer may try to put data into buffer when it is full, and Consumer may try to remove data from buffer when it is empty.

Solution: The solution to this problem is: The producer either sleep or discard the data when the buffer is full and Consumer sleep when it finds buffer empty. Once the producer puts the data in the buffer, he wakes up the sleeping consumer similarly, once the consumer removes the data from buffer, he wakes up sleeping producer, Hence, this problem is also known as **Sleep and Wake Problem**.

Dining Philosopher Problem: Consider there are five philosopher sitting around a circular table with bowls of spaghetti and five forks are placed between each pair of adjacent philosopher. At any instant, a philosopher can either eat or think and when a philosopher wants to eat, he uses two forks: one from his left and one from his right and when a philosopher thinks he puts down both the forks.

- When a philosopher wants to eat, he will wait for fork at his left and picks up when available.
- He waits for fork at his right and picks up when available.
- Start eating.
- Puts down left fork when finishes eating.
- Puts down right fork also and start thinking.

In this case, there is a situation when all the philosopher picks up fork from his left and no one can eat as fork to their right is not available. This situation is called Deadlock and leads to Starvation.

Solution: Since, all the philosophers are picking his left fork. In this solution, make one philosopher to pick right fork first.

- P_1 picks fork from his left which is F_1 and when he tries to pick his right fork which is F_2 it is already picked by P_2 as it is his left fork. So, P_1 has to wait.
- When P_2 tries to pick his right fork, which is F_3 it is already picked by P_3 as it is his left fork. So, P_2 has to wait.
- When P_3 tries to pick his right fork, which is F_4 it is already picked by P_4 as it is his left fork. So, P_3 has to wait.

- Now, P_4 has his left fork F_4 and P_5 is made pick his right fork first which is not available so, P_5 has to wait and P_4 easily get fork F_5 also.
- Now, P_4 will finish his eating and free his both forks after P_3, P_2 and P_1 will finish his eating in the same order. Finally, P_5 can also finish his eating.

Reader's Writer's Problem: There are two type of processes one is called Reader process and other is called Writer process and all the processes try to access the shared resources.

There are total four cases:

- **Case 1:** When is reader is reading the resource, a writer can't write or modify that resource at that time.
- **Case 2:** When a writer is writing or modifying the resource no reader can read the resource.
- **Case 3:** When a writer is writing or modifying the resource no other writer can modify or write the resource.
- **Case 4**: Any number of reader can read the resource simultaneously and this case doesn't create any problem.

Deadlock: Every process needs some resources to complete its execution in the following way: request a Resource, use the Resource and release the Resource

Deadlock is a situation in which a process holds a resource and waits for another resource that is blocked by another process and that process is also waiting for another resource that is blocked by some another process and all the process are blocked which is known as Deadlock.

Necessary condition for Deadlock:

- **Mutual Exclusion:** Resources are non-sharable.
- **Hold and Wait**: A process is waiting for another resource while holding a resource.
- **No Pre-emption:** Process once scheduled will be executed till its completion.
- **Circular Wait:** All the processes are waiting for some resources in a circular manner.

Methods for handling deadlock: There are for methods for handling deadlock:

1. Deadlock Ignorance: If deadlock is very rare and it happens only 1 out of 100 times, then to use deadlock handling mechanism is of no use and completely unnecessary and performance of system also decreases if deadlock handling mechanism is used all the times.

2. Deadlock Prevention: Deadlock occurs only when one of the four necessary conditions hold simultaneously and if by any means we can violate any one condition, then deadlock will never occur.

- **Mutual Exclusion:** It can be avoided if resources are shared among processes but some resources like printers and tape drivers are non-sharable. So, mutual exclusion can't be avoided completely.
- **Hold and Wait:** Try to give the all the required resources to process before the start of execution. But this will lead to **low device utilization** as the resource which is required for less time is blocked till the completion of execution.
- **No Pre-emption:** No pre-emption can be avoided by using time quantum or priority.
- **Circular Wait:** All the resources are given some priority number and a process can use those resources in either increasing or decreasing order of priority.

3. Deadlock Avoidance: Deadlock avoidance is a mechanism in which deadlock is avoided by allocating resources to a process only when it doesn't lead to deadlock. Each process must declare the maximum number of resources it needs. A state where no deadlock occurs is called **Safe State**, conversely a deadlock state is known as **Unsafe State**.

Banker's Algorithm to avoid deadlock: Banker's algorithm is a resource allocation and deadlock avoidance algorithm. Inputs to this algorithm are maximum number of resources needed by a processor, maximum freely available resources and currently blocked resources by each process. Request for resources by any process is granted only when the requested resources are less than both maximum freely available resources and maximum needed resources by that process.

4. Deadlock Detection and Recovery: For deadlock detection and recovery, Resource Allocation Graph (RAG) is used. So, we must learn about RAG first.

Resource Allocation Graph (RAG): It is a pictorial representation in which processes are represented using circle and resources are represented using a rectangle.

Resources allocation to a process is shown by an arrow in which head of arrow is attached to process and tail of the arrow is attached to resource. Resource request is shown by an arrow in which tail of arrow is attached to process and head of the arrow is attached to resource.

When all the resources have single instance of resources and any cycle is formed in RAG, then the system is in deadlock state. When all the resources have more than one instances of resources, then cycle in RAG is not sufficient condition to detect deadlock in the system.

The OS detects the deadlock in the system by using RAG and perform the recovery operation by considering either processes or resources:

- **By process killing:** By killing a single process or all the processes that causes deadlock. While killing a single process, OS chooses that process to kill which has done least amount of work.
- **Pre-empt the resource and rollback to a safe state:** Resources are pre-empted from a deadlock causing process and allocated to other process. OS keeps track of every state and in case of deadlock, OS can roll back the system to a previous safe state.
- **Starvation and Aging:** Starvation occurs when a process doesn't get the required resources for a long time. **Aging** is technique to avoid starvation in the system by using aging factor called priority. It increases the priority of a process depending on the waiting time. As the waiting time of a process increases, aging factor (priority) of that process will be also increases.

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Lesson 3: Database Management System

Database Management System: Database is collection of data/items/fields which contains information and DBMS is a set of interrelated data and a group of programs (or set of rules) to access those data. DBMS helps efficient retrieval, insertion, manipulation and deletion of data from database.

Database System Application: Databases are widely used. Here are some representative applications: Finance, Universities, Airlines, Human resources, Banking, Manufacturing and Sales.

File-processing system: Stores permanent records in various files, and it needs different application programs to extract records from, and add records to, the appropriate files.

Application Architectures: Users of a database system do not use the database directly but connect to the database through a network. Users work on client machine and database system runs on server machine. Application architecture is of two types:

1. Two-tier architecture: It is similar to a basic client-server model. In this, the application is divided into a module that resides at the client machine, which communicate directly to the database system at the server machine through query language commands.

2. Three-tier architecture: In this architecture, the client machine does not communicate with database system directly. Instead, the client end send request to an application server, through a network which in turn communicates with a database system to access data. This type of model prevents direct interaction of the client with the server thereby reducing unauthorized access to data. Hence, it is more secure.

Data Abstraction: There are three level of data abstraction in database systems.

1. Physical Level: The lowest level of abstraction describes how the data is actually stored. It is also known as Internal level. For example- student, customer, employee, account records.

2. Logical/Conceptual Level: The next-higher level of abstraction describes what data are stored in the database, and what relationships exist among those data.

3. View/External Level: The highest level of abstraction describes the database part that a particular user group is interested and hides the remaining database from that user group. The view level of abstraction exists to simplify their interaction with the database system.

Instances and Schemas:

- The group of data/information stored in the database at a particular point of time is called an **Instance** of the database.
- The logical structure of database is called the database **schemas**.
- The **physical schema** describes the database design at the physical level.
- The **logical schema** describes the database design at the logical level.
- The **view** or **subschema** describe the database design at the view level.
- The ability to modify physical schema without changing the logical schema is called as **Physical Data Independence**.

Data Model: It is a collection of conceptual tools for describing data, relationship of data, data semantics and consistency constrains. There are many types of data models:

1. Entity Relationship (E-R) data Model: It is based on a notion of a real-world objects, called entities, and relationships among these objects. It defines the conceptual view of the database and represent the database as a diagram.

Entity Relationship Diagram: An Entity Relationship (E-R) diagram can describe graphically the overall conceptual or logical structure of a database. It consists of the following components as given in the below table: